

PCRF Transformer Reliability Executive Report

Causal Reliability Flow & Derivative-Weighted Diagnostics Dashboard for QWEN/QWEN2.5-0.5B

1. Executive Summary & Core Gating Status

- **Target Architecture Profile:** QWEN/QWEN2.5-0.5B
- **Automated Path C Gating (validation fallback gating):** DISABLED (Discrete Exact Match accuracy utilized)
- **Composite PCRF Adoption Index:** 45.06 / 100
- **Strategic Deployment Directive:** REJECT ADOPTION: RESTORE BASELINE CONFIGURATIONS
- **Production Security Risk Level:** ● [HIGH OVER-STEERING RISK | BLOCK PARAMETER MUTATIONS]

System Reliability Metrics At-a-Glance

Performance Parameter	Baseline (Pre-Intervention)	Post-Regularization	Net Gain / Loss Delta
Seen Validation Accuracy	45.00%	50.00%	+5.00%
Unseen Generalization Acc	45.00%	50.00%	+5.00%
Seen Validation Loss (NLL)	4.22905	3.66871	-0.56034
Unseen Validation Loss (NLL)	4.21506	3.89519	-0.31987
Unseen Perplexity (PPL)	67.70	49.17	-18.53

2. Integrated PCRF Scoreboard

Feature Track / Module	Baseline Value	PCRF Result Value	Track Score	Gating Status
Derivatives	0.00 (Unmeasured)	0.00364 (Avg Sensitivity)	3.6/100	SAFE_TO_APPLY
Curriculum Curation	Uniform Selection (Std=0.0)	PCRF Prioritized (Std=3.70)	74.0/100	SAFE_TO_APPLY
Structural Depth Monitor	Unmonitored Depth	Chain Reliability: 38.47%	38.5/100	MEASUREMENT_ONLY
Safe SFT Regularization	Unseen Acc: 45.0%	Unseen Acc: 50.0%	60.0/100	DO_NOT_APPLY

3. Deep-Dive Diagnostics: Structural Bottlenecks & Representation Decay

Dynamic Decay Curve Insights for QWEN/QWEN2.5-0.5B

Causal analysis on QWEN/QWEN2.5-0.5B revealed the following trends across its structural segments:

- **Calculated System Chain Reliability (R_{sys}):** 38.4660%
- **Triggered Structural Flaw Heuristics:** RESIDUAL_STREAM_BYPASS_DETECTED

Layer-wise Survival & Birnbaum Sensitivity Index (mathematical sensitivity expressing downstream changes)

Layer	Survival Probability (r_l)	Birnbaum Derivative (D_R)
Block 00	100.00%	0.48721
Block 01	100.00%	0.45565
Block 02	100.00%	0.39047
Block 03	100.00%	0.38601
Block 04	100.00%	0.38589
Block 05	100.00%	0.38774
Block 06	100.00%	0.38762
Block 07	100.00%	0.38910
Block 08	100.00%	0.38960
Block 09	100.00%	0.38947
Block 10	100.00%	0.38836
Block 11	100.00%	0.38811
Block 12	100.00%	0.38811
Block 13	100.00%	0.38836
Block 14	100.00%	0.38985
Block 15	100.00%	0.39010
Block 16	100.00%	0.39084
Block 17	100.00%	0.39447
Block 18	100.00%	0.39624
Block 19	100.00%	0.40056
Block 20	100.00%	0.40752
Block 21	100.00%	0.52731
Block 22	100.00%	0.54597
Block 23	100.00%	0.65482

- **Dynamic Layer Integrity Trend Analysis:** Flat Representational Stability Pattern
 - Input Segment Average (First 15%): 100.00%
 - Highway Segment Average (Middle 70%): 100.00%
 - Output Segment Average (Last 15%): 100.00%

4. Empirical Perturbation Analysis: Layer-wise Sensitivity

- **Average System Sensitivity to Perturbation:** 0.00331
- **Highest Sensitivity Bottleneck Layer:** Layer 1 (Max Drift Delta: 0.01200)

Layer Index	Clean Target Prob	Perturbed Target Prob	Delta Prob
Layer 0	11.94%	10.89%	0.01049
Layer 1	11.94%	10.74%	0.01200
Layer 2	11.94%	11.49%	0.00457
Layer 3	11.94%	11.64%	0.00303
Layer 4	11.94%	11.53%	0.00417
Layer 5	11.94%	11.61%	0.00327
Layer 6	11.94%	12.69%	-0.00746
Layer 7	11.94%	11.96%	-0.00018
Layer 8	11.94%	12.00%	-0.00060
Layer 9	11.94%	11.51%	0.00435
Layer 10	11.94%	12.08%	-0.00140
Layer 11	11.94%	11.68%	0.00263
Layer 12	11.94%	12.49%	-0.00547
Layer 13	11.94%	11.92%	0.00021
Layer 14	11.94%	11.65%	0.00290
Layer 15	11.94%	11.96%	-0.00022
Layer 16	11.94%	11.55%	0.00394
Layer 17	11.94%	11.57%	0.00370
Layer 18	11.94%	11.95%	-0.00009
Layer 19	11.94%	11.58%	0.00367
Layer 20	11.94%	11.78%	0.00163
Layer 21	11.94%	11.78%	0.00166
Layer 22	11.94%	11.97%	-0.00029

Layer Index	Clean Target Prob	Perturbed Target Prob	Delta Prob
Layer 23	11.94%	11.80%	0.00144

5. Hallucination Risk & Output Confidence

The PCRF framework implements deep diagnostics of decoding confidence to guarantee **Calibrated Ignorance** (i.e. model is less confident when uncertain, rather than generating high-confidence wrong answers). By regularizing boundary layer representation alignments via CDL v2, the system directly controls downstream confidence collapses on incorrect outputs.

Hallucination Prevention Audit Scorecard

Diagnostic Metric	Measured Count	Engineering Definition & Protective Scope
Total Baseline Hallucinations Found	22	Validation prompts where the baseline model failed to match ground-truth.
Active Hallucination Repairs Promoted	2	Baseline errors cleanly aligned and repaired in the candidate model.
Candidate Over-Steers Prevented	19	Both models were wrong, but candidate risk was higher; router fell back to baseline.
Catastrophic Regressions Blocked	0	Baseline was correct but candidate failed; actively blocked in production.
Net Gateway Interventions	21	Overall cases actively guarded by the Protected Router (100% active coverage).

"Calibrated Ignorance" Trace Showcase

CASE 1: Prompt ID 119

- Prompt:** Complete with one word only: A formal logical interface allowing separate software modules to interact is an
- Expected Target:** API
- Baseline Output:** example of a(n) _____ | **Candidate Output:** example of a(n) _____
- Telemetry Detection:** System reliability R_{sys} collapsed to 0.3847. Representational instability triggered late-layer entropy spikes.
- CDL v2 Action:** Collapsed candidate's Top-1 probability from 53.91% down to 48.63%.
- Gateway Action:** HallucinationFound: "example of a(n) _____" -> [Gateway Router Override: Fallback to baseline due to representational instability (Risk Score: Candidate 0.3604 > Baseline 0.5629)]

CASE 2: Prompt ID 120

- Prompt:** Complete with one word only: The physical block boundary used to serialize hard drive data tracks is a
- Expected Target:** Sector

- **Baseline Output:** (n) _____. A | **Candidate Output:** (n) _____. A
- **Telemetry Detection:** Late-layer structural entropy S_l spiked. Cosine representation similarity dropped below baseline limits.
- **CDL v2 Action:** Successfully suppressed the formatting token probability, collapsing it from 9.72% to 4.47%.
- **Gateway Action:** HallucinationFound: "(n) _____. A" -> [Gateway Router Override: Fallback to baseline due to representational instability (Risk Score: Candidate 0.4443 > Baseline 0.4062)]

CASE 3: Prompt ID 110

- **Prompt:** Complete with one word only: The dense celestial body whose localized gravitational path traps light is a
- **Expected Target:** Black Hole
- **Baseline Output:** _____. A. | **Candidate Output:** black hole. The dense celestial
- **Telemetry Detection:** Mid-to-late highway transitions failed to match structural targets (R_{sys} degradation).
- **CDL v2 Action:** Extinguished pre-training option headers. Candidate confidence lowered from 10.06% to 9.18%.
- **Gateway Action:** HallucinationFound: "black hole. The dense celestial" -> Repair Approved: Candidate successfully aligned prompt target.

6. Curriculum Prioritization & Semantic Error Concentration

The curriculum prioritization scoring system matches training prompts with a priority score:

Priority Score = $NLL \times \left(1.0 + \sum_l \Delta_l\right)$

This mathematical approach isolates low-impact prompts and prioritizes abstract syntax and logical structures. For example, QWEN/QWEN2.5-0.5B consistently struggles with advanced computer science and coding syntax prompts, while basic factual lookups remain stable.

Top 5 Highest Cascade-Risk Training Prompts

ID	Prompt	Target Completion	Priority Score
79	Complete with one word only: The reserved programming keyword used to initiate routine blocks in Python is	def	16.37
66	Complete with one word only: The digital counting framework representing information with 0 and 1 is	Binary	16.31
78	Complete with one word only: The reserved programming keyword used to declare structural blueprints in Python is	class	16.27
56	Complete with one word only: A liquid solution with a pH rating significantly higher than 7 is a	Base	15.64

ID	Prompt	Target Completion	Priority Score
74	Complete with one word only: The relational database directive used to fetch selected tuples from table arrays is	SELECT	15.34

Bottom 5 Lowest Cascade-Risk Training Prompts

ID	Prompt	Target Completion	Priority Score
40	Complete with one word only: The official capital city of Indonesia is	Jakarta	3.87
20	Complete with one word only: The official capital city of Thailand is	Bangkok	3.87
5	Complete with one word only: The official capital city of Japan is	Tokyo	3.63
29	Complete with one word only: The official capital city of Ireland is	Dublin	3.61
16	Complete with one word only: The official capital city of Argentina is	Buenos Aires	3.38

7. Root-Cause Debugging Hub: Failed Generations

The following table records the precise prompts where the model failed to generate an exact semantic match during validation evaluation. Use this trace to identify and rectify model logic regressions.

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
seen_val	81	Complete with one word only: The official capital city of So...	Seoul	`_____.		
A.`		Seoul. Seoul is the official	1.6797			
seen_val	84	Complete with one word only: The official capital city of Sw...	Bern	`_____.		

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
A.`	`_____.					
A.`	3.5469					
seen_val	86	Complete with one word only: The noble element designated by...	Neon	`____		
A. Sodium`	`____					
A. Sodium`	9.9375					
seen_val	87	Complete with one word only: The volatile element designated...	Sulfur	`____		
A. Sodium`	`____					
A. Sodium`	3.3125					
seen_val	89	Complete with one word only: The yellow dwarf star supportin...	Sun	closest star to Earth, and	closest star to Earth, and	3.2031
seen_val	90	Complete with one word only: Mechanical acoustics are comple...	Vacuum	medium. They are only able	medium. They are only able	12.5000
seen_val	96	Complete with one word only: To enforce unique constraints w...	Set	technique called "hashing."	technique called hashing. Hashing	6.5625
seen_val	97	Complete with one word only: The hypermedia syntax used to f...	HTML	`called _____.		

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
A`	`called _____.					
A`	3.8750					
seen_val	98	Complete with one word only: An execution failure originatin...	Bug	`(n) _____.		
A`	`(n) ____.					
A.`	10.6875					
seen_val	99	Complete with one word only: A standardized text notation re...	JSON	`called a ____.		
A.`	`called a ____.					
A.`	10.4375					
...	...	...	...	...	...	(And 12 more trace details)

8. Actionable 4-Phase Enterprise Upgrade Strategy for QWEN/QWEN2.5-0.5B

Phase 1: Selective SFT Boundary Layer Regularization

Stop fine-tuning all model weights. Freeze intermediate layers. Concentrate compute budgets on the high-risk boundary layers (Layers 0-1 and late projection blocks) to shrink memory footprints while protecting historical parameter arrays.

Phase 2: Automated PCRF Dataset Curation

Deploy the `curriculum_scores.csv` output as a filtration gate. Discard the bottom 30% of low-priority, high-redundancy training samples to save GPU compute hours.

Phase 3: Live Production "Canary" Monitors

Deploy the continuous Structural PCRF Plugin as an active endpoint wrapper. If R_{sys} collapses below 75% for a given user query, route that query to a safe fallback model before a hallucinated output reaches the user.

Phase 4: Automated CI/CD Governance Safety Gates

Integrate the Safe PCRF Controller as a hard gate in your Git-Ops pipelines. No model should ever be promoted unless it satisfies the continuous Path C non-inferiority constraints.

9. Compute Environment & Host Profile Audit

- **Host Platform:** Linux 6.1.0-49-cloud-amd64
- **Detected CPU Cores:** 8
- **Host Memory Capacity:** 29.38 GB
- **GPU Platform Context:** Tesla T4 (14.56 GB VRAM)

Report programmatically generated by PCRF Reliability Suite v1.